Quinn, J., Lee, S., Greeley, D., Gehman, A., Kuris, A. M., & Wood, C. L. (2021). Long-term change in the parasite burden of shore crabs (*Hemigrapsus oregonensis* and *Hemigrapsus nudus*) on the northwestern Pacific coast of North America. *Proceedings of the Royal Society B: Biological Sciences*, 288(1945), 20203036. https://doi.org/10.1098/rspb.2020.3036

This paper investigated whether populations have shifted over time in two shore crabs. Data collected suggests that the abundance of the parasitic isopod Porunion Conformis has been stable for the past 50 years. However, larval microphallid trematodes increased in prevalence over time among H. oregonensis hosts, from a mean of 8.4-61.8%.

- 3 parasite taxa were observed: P. conformis, larval cystacanthus (acanthocephalans) , larval metacercariae (trematodes)
- 36% of all crabs sampled were infected with P. conformis
- P. conformis did not change in abundance over time in H. oregonensis

Watts, A. J. R., Lewis, C., Goodhead, R. M., Beckett, S. J., Moger, J., Tyler, C. R., & Galloway, T. S. (2014). Uptake and retention of microplastics by the shore crab *Carcinus maenas*. *Environmental Science & Technology*, 48(15), 8823–8830. https://doi.org/10.1021/es501090e

This paper investigated the uptake, location, residence time, and the clearance of microplastics polystyrene microspheres, 8-10um in diameter in Carcinus maenas by direct ventilatory exposure and exposure through consumption by the common mussel. Ingested microspheres were retained within the body tissues of crabs for up to 14 days following ingestion and up to 21 days. Inspiration across the gill was significantly higher into the posterior versus anterior gills.

- The gut is divided into fore, mid and hindgut. The foregut contains complex gastric mill that grind and mix plant and animal tissue. Particles >100 nm pass to the hindgut and are eliminated in feces. Particles <100nm pass to the hepatopancreas for digestion
- Major function of the posterior gills is ion exchanges, while the anterior gills serve as the major site for respiration and oxygen uptake

Birch, A., Cox, K. D., Murchy, K. A., Emry, S., & Harley, C. D. G. (2025). Shipping noise tolerance in invertebrates: A case study of the shore crab *Hemigrapsus oregonensis*. *PLOS ONE*, *20*(8), e0329098. https://doi.org/10.1371/journal.pone.0329098

This paper investigated the behavioral impacts of shipping noise on H. oregonensis and whether they can develop tolerance to noise. Results indicate that ship noise significantly impacts shore crab's initial response after a simulated predator attack. However, ship noise did not significantly impact whether the crabs retreated to shelter after a predator attack, nor did it disrupt feeding. The interaction between treatment and site type indicates there is no evidence of tolerance related to prior noise exposure

- Research indicates that setae, chordontal organs, and statocysts within the antennules are all capable of detecting particle motion caused by sound vibrations
- Antennae are believed to be involved in sound perception
- Crustaceans are less sensitive to the pressure component of sound but are more responsive to lower frequency sounds
- Shipping noise can increase oxygen consumption, indicating higher metabolic rates and potential elevated stress levels
- Consistent with broader literature which suggests that marine noise influences arthropod behavior can be positive, negative or null

Kuris, A. M., Poinar, G. O., & Hess, R. T. (1980). Post-larval mortality of the endoparasitic isopod castrator Portunion conformis (Epicaridea: Entoniscidae) in the shore crab, Hemigrapsus oregonensis, with a description of the host response. Parasitology, 80(2), 211–232. doi:10.1017/S003118200000069X
https://www.cambridge.org/core/journals/parasitology/article/postlarval-mortality-of-the-endoparasitic-isopod-castrator-portunion-conformis-epicaridea-entoniscidae-in-the-shore-crab-hemigrapsus-oregonensis-wit

Hemigrapsus oregonensis sometimes kill the female Portunion Conformis. Hosts with dead P.conformis and total parasite population found dead are associated with high prevalence of 1. High prevalence of parasitism, 2. Female hosts, and 3. Large hosts. The parasite's death is typically the result of activation of a successful host defensive process rather than indicative of a defect on the part of the parasites. Parasitized female hosts can regain their reproductive capabilities following the death of the parasite, but broods are smaller.

- Dead female entoniscides are sometimes found encapsulated or degenerating within host crabs

Moksnes P, Pihl L, Montfrans Jv (1998) Predation on postlarvae and juveniles of the shore crab Carcinus maenas: importance of shelter, size and cannibalism. Mar Ecol Prog Ser 166:211-225
https://doi.org/10.3354/meps166211

This paper investigates the relationships in postlarvae and early juvenile stages of Carnicus maenas by studying the habitat and size related habitat mortality. Results indicate that predation is an important process that can create a bottleneck for juvenile shore crab populations during settlement and early juvenile stages. Post larvae obtained refuge from predation in several different habitats, suggesting that the recruitment of juvenile shore crabs will be less affected by temporal and spatial variation of any single habitat. Cannabialistic juveniles caused predation rates similar to or higher than other investigated predators on postlarvae and young juvenile crabs in nursery areas.

- Cannibalistic juvenile shore crabs and brown shrimp caused 90 and 80% mortality respectively
- Predation caused significant mortality in all habitats, but blue mussels eelgrass and filamentous algae (best habitat) provided a significant refuge from predation compared to sandy habitats

Dawirs, R. R. (1985). Temperature and larval development of Carcinus maenas (Decapoda) in the laboratory; predictions of larval dynamics in the sea. *Marine Ecology Progress Series*, *24*(3), 297–302.
http://www.jstor.org/stable/24816895

Larvae of Carcinus maenas were reared in the laboratory from hatching to metamorphosis at different temperatures. This paper investigated 1. To what extent is the duration of larval development influenced by water temperature 2. If it is possible to predict the occurrence and presence of subsequent larval stages in nature relative to the date of hatching. Results show that larvae were reared successfully at constant water temperature between 12 and 25c and that prediction is possible?

- Larvae which hatch early in the year need more time to metamorphose than later larvae due to increasing temperatures

Symons, P. E. K. (1964). Behavioral responses of the crab Hemigrapsus oregonensis to temperature, diurnal light variation and food stimuli. Ecology, 45(3), 431–673.

In this paper, H. Oregonensis were reared between 5-18c for 8 days and observed for 8 days. From this experiment researchers found:

- Crabs reared at higher temperature exhibited higher movement. With males traveling farther than females
- Crabs held at 5C traveled almost identical distances as crabs held at 7 and 11c.
- A sudden drop in temperature, from 18c to 10c significantly decreased the distance traveled.
- Even in the absence of food, stimuli, feeding movements, probing movements, and distance traveled showed trends to increase with progressive starvation,

Anger, K., Spivak, E., & Luppi, T. (1998). Effects of reduced salinities on development and bioenergetics of early larval shore crab, Carcinus maenas. Journal of Experimental Marine Biology and Ecology, 220(2), 287–304.

In this study, researchers investigated hypo-osmotic survival, development, metabolism, and growth in early larvae of C.maenas reared at four different salinities. From this experiment, researchers found that Zoea I shore crab larvae react during the late stages of their moulting cycle less sensitively against reduced salinities than during postmoult and intermoult

- Crab reared at 15% did not allow for development beyond the first zoel stage
- Metamorphosis to the megalopa was reached at salinities >20, but was significantly delayed and mortality enhanced as compared with 25 and 32?
- Rates of growth and respiration decreased during exposure to reduced salinities <25%
- Significant deceases were found in the rates of early zoel development, growth, respiration, and assimilation

Whatcom Watch. (2016, October). Invasive Green Crabs Found in Padilla Bay. Whatcom Watch Online. https://whatcomwatch.org/index.php/article/invasive-green-crabs-found-in-padilla-bay/

Off the shore of padilla bay, four green crabs have been caught, indicating an early sign of an invasion. This is a problem as European green crabs can have detrimental effects on the habitat and on native species. They feed on young dungeness and other crabs before they flee into deeper water where they burrow into mud or clay banks where they can reduce the number of clams and worms in the sediment. In many places they have reduced shellfish population, uprooting eelgrass which is an important habitat to other crabs and fish.

Robitzski, D. (2017). Prozac Puts Crabs in a Mood to Take Deadly Risks. LiveScience https://www.lives5rcience.com/60766-crabs-on-prozac-take-deadly-risks.html

Researchers exposed crabs to small amounts of fluoxetine and observed their behaviors under laboratory conditions. Over time, H. Oregenensis drugged with prozac have been shown to take elevated risky behaviors such as foraging during the day, foraging in the presence of a predator, and fighting each other. Crabs living around estuaries and harbors have increased contact with fluoxetine by contaminated runoff which will result in greater risk of predation and mortality.